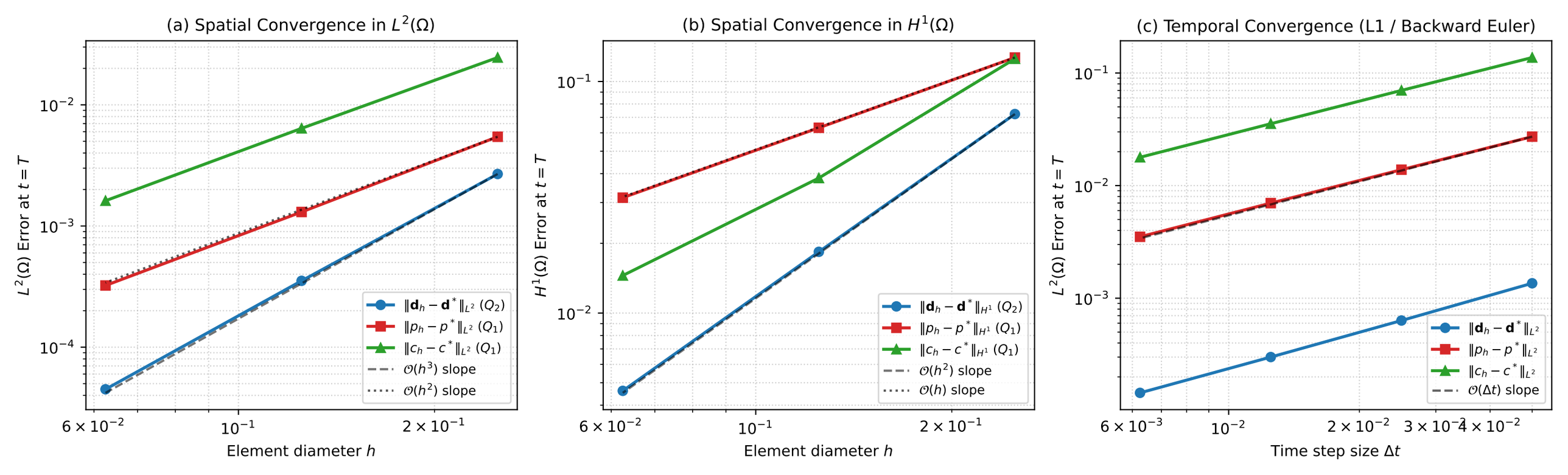
(a) Spatial Convergence in $L^2(\Omega)$
$L^2(\Omega)$ Error at $t = T$
Element diameter h
$\|\mathbf{d}_h - \mathbf{d}^*\|_{L^2}$ (Q_2)
$\|p_h - p^*\|_{L^2}$ (Q_1)
$\|c_h - c^*\|_{L^2}$ (Q_1)
$\mathcal{O}(h^3)$ slope
$\mathcal{O}(h^2)$ slope
(b) Spatial Convergence in $H^1(\Omega)$
$H^1(\Omega)$ Error at $t = T$
Element diameter h
$\|\mathbf{d}_h - \mathbf{d}^*\|_{H^1}$ (Q_2)
$\|p_h - p^*\|_{H^1}$ (Q_1)
$\|c_h - c^*\|_{H^1}$ (Q_1)
$\mathcal{O}(h^2)$ slope
$\mathcal{O}(h)$ slope
(c) Temporal Convergence (L1 / Backward Euler)
$L^2(\Omega)$ Error at $t = T$
Time step size Δt
$\|\mathbf{d}_h - \mathbf{d}^*\|_{L^2}$
$\|p_h - p^*\|_{L^2}$
$\|c_h - c^*\|_{L^2}$
$\mathcal{O}(\Delta t)$ slope